



MATH130: Polynomials and Factoring

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Learning Objectives

- Apply the laws of exponents (product, quotient, power)
- Multiply polynomials using FOIL and the distributive property
- Factor polynomials using GCF, trinomial, and special patterns

Simplify each expression completely. Show all steps and circle your final answer.

Laws of exponents — product rule

1. Simplify: $x^7 \cdot x^1$. Write your answer as x^k .

$$x^7 \cdot x^1$$

Answer: _____

2. Simplify: $x^2 \cdot x^4$. Write your answer as x^k .

$$x^2 \cdot x^4$$

Answer: _____

3. Simplify: $x^5 \cdot x^3$. Write your answer as x^k .

$$x^5 \cdot x^3$$

Answer: _____

4. Simplify: $x^3 \cdot x^4$. Write your answer as x^k .

$$x^3 \cdot x^4$$

Answer: _____

5. Simplify: $x^6 \cdot x^2$. Write your answer as x^k .

$$x^6 \cdot x^2$$

Answer: _____

6. Simplify: $x^6 \cdot x^2$. Write your answer as x^k .

$$x^6 \cdot x^2$$

Answer: _____

7. Simplify: $x^2 \cdot x^3$. Write your answer as x^k .

$$x^2 \cdot x^3$$

Answer: _____

8. Simplify: $x^2 \cdot x^3$. Write your answer as x^k .

$$x^2 \cdot x^3$$

Answer: _____

Laws of exponents — quotient rule

9. Simplify: x^4 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^4 \div x^1$$

Answer: _____

10. Simplify: x^4 / x^3 (assume $m > n$). Write your answer as x^k .

$$x^4 \div x^3$$

Answer: _____

11. Simplify: x^7 / x^2 (assume $m > n$). Write your answer as x^k .

$$x^7 \div x^2$$

Answer: _____

12. Simplify: x^3 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^3 \div x^1$$

Answer: _____

13. Simplify: x^3 / x^2 (assume $m > n$). Write your answer as x^k .

$$x^3 \div x^2$$

Answer: _____

14. Simplify: x^8 / x^4 (assume $m > n$). Write your answer as x^k .

$$x^8 \div x^4$$

Answer: _____

15. Simplify: x^2 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^2 \div x^1$$

Answer: _____

16. Simplify: x^8 / x^3 (assume $m > n$). Write your answer as x^k .

$$x^8 \div x^3$$

Answer: _____

Multiplying polynomials (FOIL)

17. Multiply: $(4x + 5)(1x + 2)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 5)(1x + 2)$$

Answer: _____

18. Multiply: $(x + 4)(x - 4)$. Write the result in standard form.

$$(x + 4)(x - 4)$$

Answer: _____

19. Multiply: $(3x + 2)(3x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 2)(3x + 5)$$

Answer: _____

20. Multiply: $(x + 2)(x - 3)$. Write the result in standard form.

$$(x + 2)(x - 3)$$

Answer: _____

21. Multiply: $(3x + 1)(4x + 3)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 1)(4x + 3)$$

Answer: _____

22. Multiply: $(x + 6)(x - 7)$. Write the result in standard form.

$$(x + 6)(x - 7)$$

Answer: _____

23. Multiply: $(1x + 4)(3x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(1x + 4)(3x + 5)$$

Answer: _____

24. Multiply: $(x + 2)(x - 3)$. Write the result in standard form.

$$(x + 2)(x - 3)$$

Answer: _____

25. Multiply: $(3x + 3)(2x + 3)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 3)(2x + 3)$$

Answer: _____

26. Multiply: $(x + 5)(x - 2)$. Write the result in standard form.

$$(x + 5)(x - 2)$$

Answer: _____

27. Multiply: $(4x + 1)(4x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 1)(4x + 5)$$

Answer: _____

28. Multiply: $(x + 5)(x - 3)$. Write the result in standard form.

$$(x + 5)(x - 3)$$

Answer: _____

29. Multiply: $(4x + 6)(1x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 6)(1x + 5)$$

Answer: _____

30. Multiply: $(x + 7)(x - 1)$. Write the result in standard form.

$$(x + 7)(x - 1)$$

Answer: _____



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ANSWER KEY & SOLUTIONS

Topics: Laws of exponents — product rule, Laws of exponents — quotient rule, Multiplying polynomials (FOIL). All answers verified by independent computation.

Solutions

Laws of exponents — product rule

1. Simplify: $x^7 \cdot x^1$. Write your answer as x^k .

$$x^7 \cdot x^1$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^7 \cdot x^1 = x^{(7+1)} = x^8.$$

Answer: x^8

2. Simplify: $x^2 \cdot x^4$. Write your answer as x^k .

$$x^2 \cdot x^4$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^2 \cdot x^4 = x^{(2+4)} = x^6.$$

Answer: x^6

3. Simplify: $x^5 \cdot x^3$. Write your answer as x^k .

$$x^5 \cdot x^3$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^5 \cdot x^3 = x^{(5+3)} = x^8.$$

Answer: x^8

4. Simplify: $x^3 \cdot x^4$. Write your answer as x^k .

$$x^3 \cdot x^4$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^3 \cdot x^4 = x^{(3+4)} = x^7.$$

Answer: x^7

5. Simplify: $x^6 \cdot x^2$. Write your answer as x^k .

$$x^6 \cdot x^2$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^6 \cdot x^2 = x^{(6+2)} = x^8.$$

Answer: x^8

6. Simplify: $x^6 \cdot x^2$. Write your answer as x^k .

$$x^6 \cdot x^2$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^6 \cdot x^2 = x^{(6+2)} = x^8.$$

Answer: x^8

7. Simplify: $x^2 \cdot x^3$. Write your answer as x^k .

$$x^2 \cdot x^3$$

→ *Product rule:* $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^2 \cdot x^3 = x^{(2+3)} = x^5.$$

Answer: x^5

8. Simplify: $x^2 \cdot x^3$. Write your answer as x^k .

$$x^2 \cdot x^3$$

→ *Product rule:* $x^m \cdot x^n = x^{(m+n)}$.

$$\rightarrow x^2 \cdot x^3 = x^{(2+3)} = x^5.$$

Answer: x^5

Laws of exponents — quotient rule

9. Simplify: x^4 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^4 \div x^1$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^4 / x^1 = x^{(4-1)} = x^3.$$

Answer: x^3

10. Simplify: x^4 / x^3 (assume $m > n$). Write your answer as x^k .

$$x^4 \div x^3$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^4 / x^3 = x^{(4-3)} = x^1.$$

Answer: x^1

11. Simplify: x^7 / x^2 (assume $m > n$). Write your answer as x^k .

$$x^7 \div x^2$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^7 / x^2 = x^{(7-2)} = x^5.$$

Answer: x^5

12. Simplify: x^3 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^3 \div x^1$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^3 / x^1 = x^{(3-1)} = x^2.$$

Answer: x^2

13. Simplify: x^3 / x^2 (assume $m > n$). Write your answer as x^k .

$$x^3 \div x^2$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^3 / x^2 = x^{(3-2)} = x^1.$$

Answer: x^1

14. Simplify: x^8 / x^4 (assume $m > n$). Write your answer as x^k .

$$x^8 \div x^4$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^8 / x^4 = x^{(8-4)} = x^4.$$

Answer: x^4

15. Simplify: x^2 / x^1 (assume $m > n$). Write your answer as x^k .

$$x^2 \div x^1$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^2 / x^1 = x^{(2-1)} = x^1.$$

Answer: x^1

16. Simplify: x^8 / x^3 (assume $m > n$). Write your answer as x^k .

$$x^8 \div x^3$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

$$\rightarrow x^8 / x^3 = x^{(8-3)} = x^5.$$

Answer: x^5

Multiplying polynomials (FOIL)

17. Multiply: $(4x + 5)(1x + 2)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 5)(1x + 2)$$

→ FOIL: First= $4 \cdot 1x^2$, Outer= $4 \cdot 2x$, Inner= $5 \cdot 1x$, Last= $5 \cdot 2$.

$$\rightarrow = 4x^2 + (8+5)x + 10.$$

$$\rightarrow = 4x^2 + 13x + 10.$$

Answer: $4x^2 + 13x + 10$

18. Multiply: $(x + 4)(x - 4)$. Write the result in standard form.

$$(x + 4)(x - 4)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (4-4)x + (4)(-4)$.

$$\rightarrow = x^2 + 0x + -16.$$

Answer: $x^2 - 16$

19. Multiply: $(3x + 2)(3x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 2)(3x + 5)$$

→ FOIL: First= $3 \cdot 3x^2$, Outer= $3 \cdot 5x$, Inner= $2 \cdot 3x$, Last= $2 \cdot 5$.

$$\rightarrow = 9x^2 + (15+6)x + 10.$$

$$\rightarrow = 9x^2 + 21x + 10.$$

Answer: $9x^2 + 21x + 10$

20. Multiply: $(x + 2)(x - 3)$. Write the result in standard form.

$$(x + 2)(x - 3)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (2-3)x + (2)(-3)$.

$$\rightarrow = x^2 + -1x + -6.$$

Answer: $x^2 - 1x - 6$

21. Multiply: $(3x + 1)(4x + 3)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 1)(4x + 3)$$

→ FOIL: First= $3 \cdot 4x^2$, Outer= $3 \cdot 3x$, Inner= $1 \cdot 4x$, Last= $1 \cdot 3$.

$$\rightarrow = 12x^2 + (9+4)x + 3.$$

$$\rightarrow = 12x^2 + 13x + 3.$$

Answer: $12x^2 + 13x + 3$

22. Multiply: $(x + 6)(x - 7)$. Write the result in standard form.

$$(x + 6)(x - 7)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (6-7)x + (6)(-7)$.

$$\rightarrow = x^2 + -1x + -42.$$

Answer: $x^2 - 1x - 42$

23. Multiply: $(1x + 4)(3x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(1x + 4)(3x + 5)$$

→ FOIL: First= $1 \cdot 3x^2$, Outer= $1 \cdot 5x$, Inner= $4 \cdot 3x$, Last= $4 \cdot 5$.

→ = $3x^2 + (5+12)x + 20$.

→ = $3x^2 + 17x + 20$.

Answer: $3x^2 + 17x + 20$

24. Multiply: $(x + 2)(x - 3)$. Write the result in standard form.

$$(x + 2)(x - 3)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (2-3)x + (2)(-3)$.

→ = $x^2 + -1x + -6$.

Answer: $x^2 - 1x - 6$

25. Multiply: $(3x + 3)(2x + 3)$. Write the result in standard form $ax^2 + bx + c$.

$$(3x + 3)(2x + 3)$$

→ FOIL: First= $3 \cdot 2x^2$, Outer= $3 \cdot 3x$, Inner= $3 \cdot 2x$, Last= $3 \cdot 3$.

→ = $6x^2 + (9+6)x + 9$.

→ = $6x^2 + 15x + 9$.

Answer: $6x^2 + 15x + 9$

26. Multiply: $(x + 5)(x - 2)$. Write the result in standard form.

$$(x + 5)(x - 2)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (5-2)x + (5)(-2)$.

→ = $x^2 + 3x + -10$.

Answer: $x^2 + 3x - 10$

27. Multiply: $(4x + 1)(4x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 1)(4x + 5)$$

→ FOIL: First= $4 \cdot 4x^2$, Outer= $4 \cdot 5x$, Inner= $1 \cdot 4x$, Last= $1 \cdot 5$.

→ = $16x^2 + (20+4)x + 5$.

→ = $16x^2 + 24x + 5$.

Answer: $16x^2 + 24x + 5$

28. Multiply: $(x + 5)(x - 3)$. Write the result in standard form.

$$(x + 5)(x - 3)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (5-3)x + (5)(-3)$.

→ = $x^2 + 2x + -15$.

Answer: $x^2 + 2x - 15$

29. Multiply: $(4x + 6)(1x + 5)$. Write the result in standard form $ax^2 + bx + c$.

$$(4x + 6)(1x + 5)$$

→ FOIL: First= $4 \cdot 1x^2$, Outer= $4 \cdot 5x$, Inner= $6 \cdot 1x$, Last= $6 \cdot 5$.

→ = $4x^2 + (20+6)x + 30$.

→ = $4x^2 + 26x + 30$.

Answer: $4x^2 + 26x + 30$

30. Multiply: $(x + 7)(x - 1)$. Write the result in standard form.

$$(x + 7)(x - 1)$$

→ FOIL with $a=1$, $c=1$: $x^2 + (7-1)x + (7)(-1)$.

→ = $x^2 + 6x - 7$.

Answer: $x^2 + 6x - 7$
